CRYPTOCURRENCY PORTFOLIO ANALYSIS

Bitcoin, Ethereum & Solana – 12 Months Performance & Risk Study

Prepared as part of the

CodeAlpha Finance & Investment Analysis Internship

Task 2: Crypto Portfolio Analysis

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Market Data As early October 2026 by Investing.com

1. Executive Summary

This report compares Bitcoin (BTC), Ethereum (ETH), and Solana (SOL), representing digital gold as a store of value, the leading smart-contract platform, and a high-performance Layer-1 challenger, respectively. Together, they account for roughly 65–70% of the total cryptocurrency market capitalization (excluding stablecoins). Despite positive structural developments over the past year, such as spot ETF approvals and inflows, major protocol upgrades, and deepening institutional adoption all three assets posted negative 12-month returns as the broader market corrected from its October 2025 highs. The magnitude of the decline scales inversely with market capitalization: Bitcoin (-28.0%) held up best, Solana (-55.0%) experienced the largest drawdown, and Ethereum (-44.0%) landed in between, directly tracking each asset's relative volatility profile.

2. Key Metrics by Asset

Bitcoin (BTC)

Bitcoin is the pioneer and largest digital asset, functioning as a decentralized macro store-of-value and serving as the primary institutional on-ramp into cryptocurrency.

$79,700 Current Price	~41% 90-Day Volatility	$1.580 Trillion Market Cap	-28.0% 1-Yr Return

52-Week Range: $57,832.50 – $126,186.00. Bitcoin trades in the lower-middle portion of its 52-week range, reflecting broader market correction while maintaining strong institutional support.

Recent News and Developments:

- Benefited from renewed spot-ETF inflows (~$4.25B over 30 days, including a single-day spike of $730.9M) and continued corporate treasury accumulation.
- Analysts like Bitwise expect Bitcoin to become structurally less volatile than mega-cap tech stocks as its institutional holder base deepens.

Ethereum (ETH)

Ethereum is the dominant smart-contract platform powering the vast majority of decentralized finance (DeFi), tokenization, and digital collectible ecosystems.

$2,500 Current Price	~60% 90-Day Volatility	$305 Billion Market Cap	-44.0% 1-Yr Return

52-Week Range: $1,388.12 – $4,955.90. Ethereum remains pressured well below its highs as it navigates scaling demands and competitive pressures from faster networks.

Recent News and Developments:

- Protocol upgrades introduced gas-abstraction features allowing users to pay transaction fees without holding native ETH directly, lowering onboarding friction for new applications.
- Spot ETH ETFs have drawn steady institutional interest despite periodic short-term outflow streaks.

Solana (SOL)

Solana is a high-throughput, low-fee Layer-1 blockchain engineered for high-frequency payments, decentralized exchanges, and consumer-scale web3 applications.

$104 Current Price	~80% 90-Day Volatility	$60.8 Billion Market Cap	-55.0% 1-Yr Return

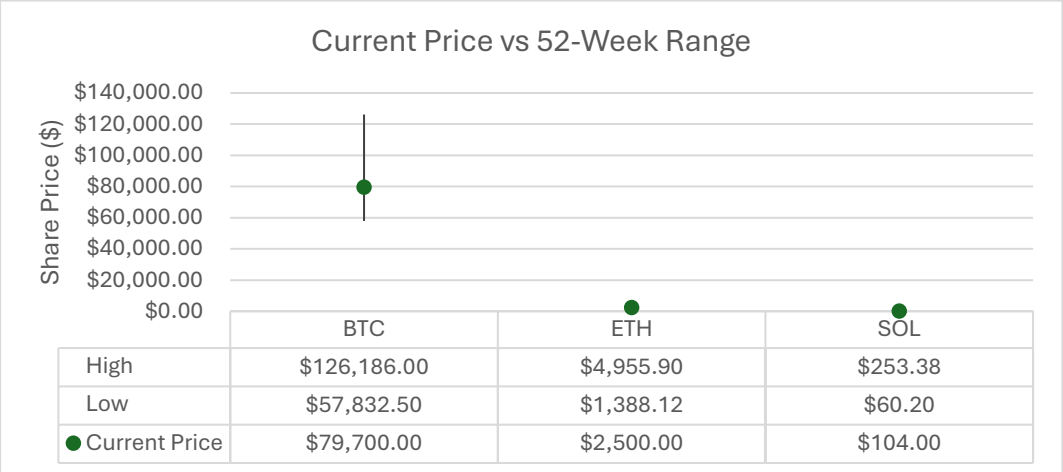
52-Week Range: $60.20 – $253.38. Solana reflects high-beta sensitivity, correcting sharply from prior peaks while sustaining robust developer and transaction momentum.

Recent News and Developments:

- The upcoming rollout of the 'Alpenglow' upgrade targets sub-second transaction finality (~150ms), positioning the network for ultra-fast settlement.
- Solana-focused investment products have seen rapid adoption, emerging as a fast-growing category for digital asset inflows alongside corporate treasury interest.

3. Visual Comparison

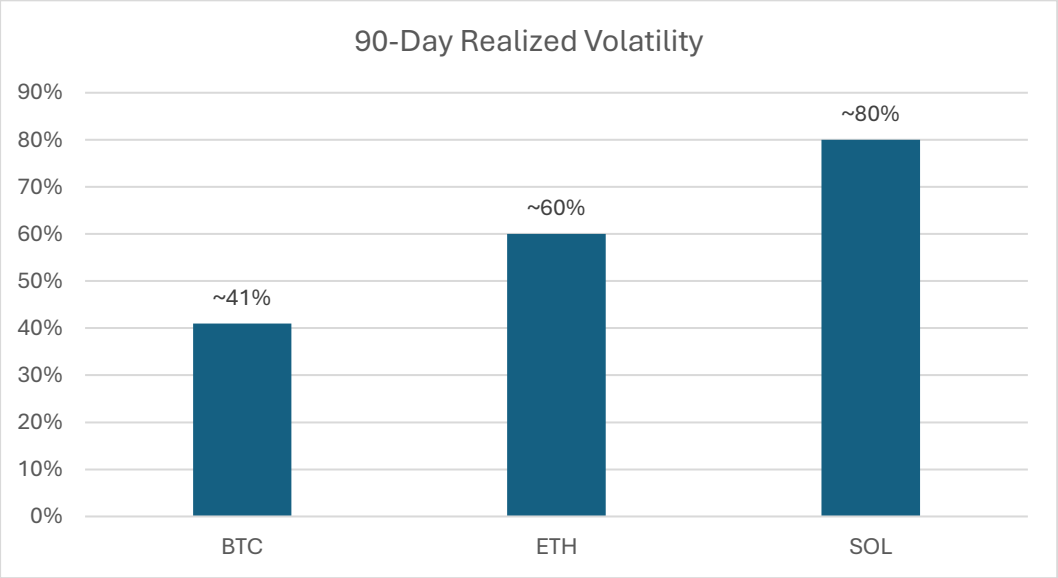
3.1 Price Position Within 52-Week Range



	BTC	ETH	SOL
High	$126,186.00	$4,955.90	$253.38
Low	$57,832.50	$1,388.12	$60.20
Current Price	$79,700.00	$2,500.00	$104.00

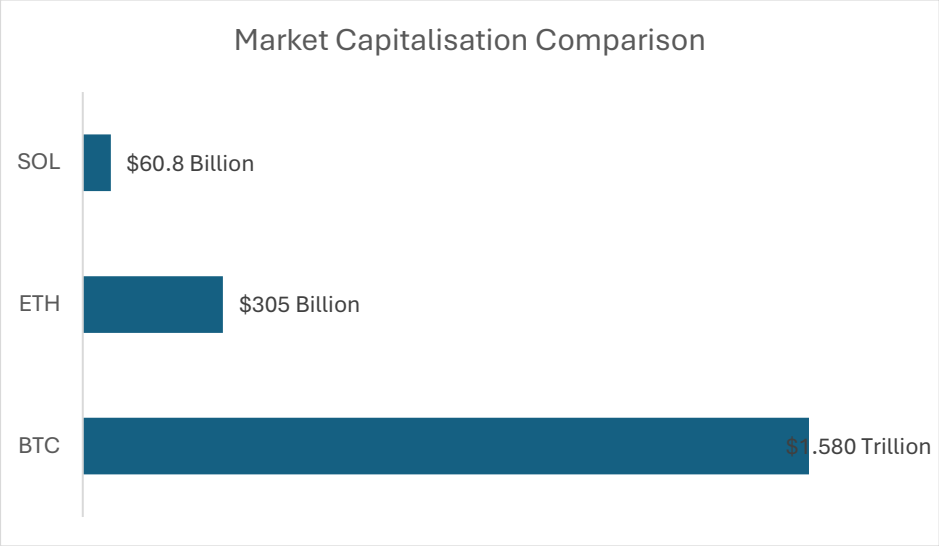
The Current Price vs. 52-Week Range chart shows that all three digital assets (Bitcoin, Ethereum, and Solana) are currently trading in the lower-to-middle third of their respective 52-week price ranges following the post-2025 market correction. Bitcoin features the highest absolute price with a 52-week low of $57,832.50 and a high of $126,186.00, currently trading at $79,700 (positioning itself at roughly 32% of its range). Ethereum trades below the midpoint of its range at $2,500, sitting at approximately 31% of its range between a low of $1,388.12 and a high of $4,955.90. Solana trades at $104, resting at about 23% of its range after rising from a 52-week low of $60.20 up to a high of $253.38.

3.2 90-Day Realized Volatility



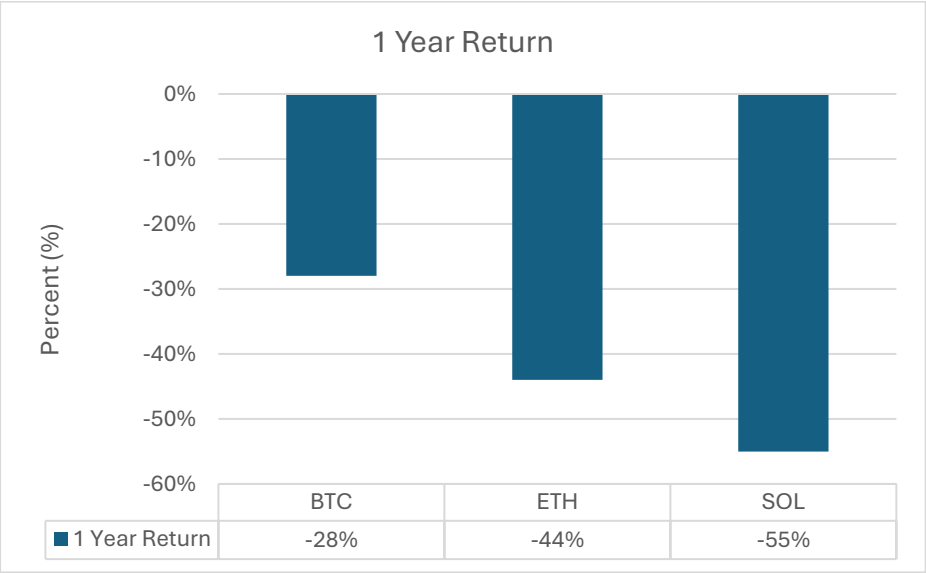
The 90-Day Realized Volatility chart shows distinct risk tiers across the three digital assets, reflecting their relative market positions. Bitcoin features the lowest volatility in the group at ~41%, underscoring its deeper liquidity and role as a macro store of value. Ethereum sits in the middle with a realized volatility of ~60%, capturing moderate price fluctuations associated with smart-contract platforms. Solana displays the highest volatility at ~80%, highlighting its high-beta profile and stronger sensitivity to market movements.

3.2 Market Capitalization



Bitcoin is by far the largest of the three ($1.580 Trillion), reflecting its dominant liquidity, institutional adoption, and role as a macro store of value. Ethereum follows as the second largest at $305 Billion, driven by its leadership in decentralized finance and smart-contract utility, while Solana rounds out the group at $60.8 Billion, underpinned by its high-throughput and low-fee Layer-1 architecture.

3.3 One-Year Price Performance



Bitcoin held up best with a 1-year return of -28%, supported by its deep liquidity and institutional adoption as a macro store of value. Ethereum experienced a moderate decline of -44%, reflecting broader market corrections while remaining the leading smart-contract platform. Solana recorded the largest drop at -55%, tracking its higher volatility and high-beta profile during the post-2025 market correction.

4. Comparative Summary Table

Metric	Microsoft (MSFT)	Apple (APPL)	NVIDIA (NVDA)
Asset Class	Digital Gold / Store of Value	Smart-Contract Platform	High-Performance Layer-1
Current Price	$79,700	$2,500	$104
Market Cap	$1.580 Trillion	$305 Billion	$60.8 Trillion
90 Day Volatility	~41%	~60%	~80%
52-Week Range	$57,832.50–$126,186.00	$1,388.12–$4,955.90	$60.20 – $253.38
1-Year Return	-28%	-44%	-55%
Risk Tier	Lower (relative)	Medium	Higher
Key Driver	Institutional inflows, treasury adoption	Gas optimization, Layer-2 scaling	Alpenglow upgrade, fast finality
Analyst Stance	Core defensive holding	Balanced smart-contract exposure	High-beta growth allocation

5. Risk Assessment

- **Bitcoin:** Lowest relative volatility and deepest institutional adoption, but still highly volatile vs. traditional assets; sensitive to Fed policy and ETF flow reversals.
- **Ethereum:** Competitive pressure from faster/cheaper L1s; recovery depends on Layer-2 ecosystem growth and sustained ETF demand.
- **Solana:** Highest volatility and drawdown risk (~80%); strong technical momentum, but position sizing should stay conservative.
- **Shared risks:** regulatory uncertainty, potential ETF outflow reversals, high sensitivity to US interest-rate expectations.

6. Conclusion and Investment Takeaways

This cryptocurrency basket demonstrates clear risk-adjusted stratification: while Bitcoin provides relative capital preservation and deep institutional liquidity, Ethereum and Solana offer exposure to high-utility smart contract networks and high-beta technological growth.

- **Conservative:** Overweight BTC (50-60% of crypto allocation) for lower relative volatility and deepest institutional backing.
- **Balanced:** 45% BTC / 30% ETH / 25% SOL (modelled in the companion Google Sheet) balances capital preservation with growth exposure.
- **Growth-oriented:** Increase allocations toward Ethereum and Solana to capture upside from protocol upgrades like Alpenglow and expanding Layer-2 ecosystems, accepting materially higher drawdown risk in exchange.